

EmotiBit: Quick Setup

Assembly

Items

EmotiBit

- **1x EmotiBit** with finger loop Emoti-stretch strap
 - Depending on your purchase, you may either have an EmotiBit MD or an EmotiBit EMO
- **1x Emoti-genic barrier** (provides an additional hygienic layer and sweat protection)
- **2x EDA electrodes (Ag/AgCl)** attached to the EmotiBit
- **2x EmotiBit stickers**

Essentials Kit

- **Adafruit Feather M0 WiFi** or **Adafruit Feather ESP32 Huzzah**
- **400mAh Lithium ion battery**
- **High-speed microSD card**
- **MicroSD card reader**
- **Micro USB cable**
- **3x Emoti-stretch straps** of different lengths to wear EmotiBit nearly anywhere on the body, ranging from a child's wrist to an adult head
- **Plastic spudger** -- used to easily toggle the hibernate switch and EmotiBit button

Electrode Kit

- **10x EDA electrodes (Ag/AgCl)**
- **4x solder-cup snaps** (to add your own EDA leads)
- **5x Emoti-genic barriers** (provides an additional hygienic layer and sweat protection)

Boards

Wi-Fi connection

1. Set up a 'config.txt' file with the following information and save it on your PC

```
{  
  "WifiCredentials": [  
    {  
      "ssid": "MY_WIFI_NAME_HERE",  
      "password": "MY_PASSWORD_HERE"  
    }  
  ]  
}
```

2. Transfer the 'config.txt' into the MicroSD card (use the USB dongle to connect to USB-A port)
3. Take out the MicroSD card from your PC and place the MicroSD reader in the EmotiBit board (as seen in Figure 1)

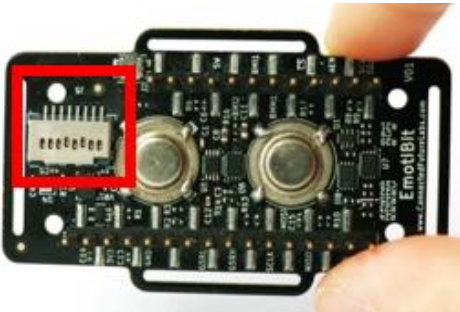


Figure 1, Insert MicroSD card into the highlighted area

Board

1. First, make sure to toggle the Hibernate switch to ON (not HIB) on the EmotiBit board



Figure 2, The HIB switch highlighted in green as shown in the EmotiBit board

2. Plug in the battery into the power connector found in the Feather and nettle the battery between the pins)
3. Stack the Feather with the EmotiBit (make sure the number of pins in the Feather matches up with the number of sockets on the EmotiBit)

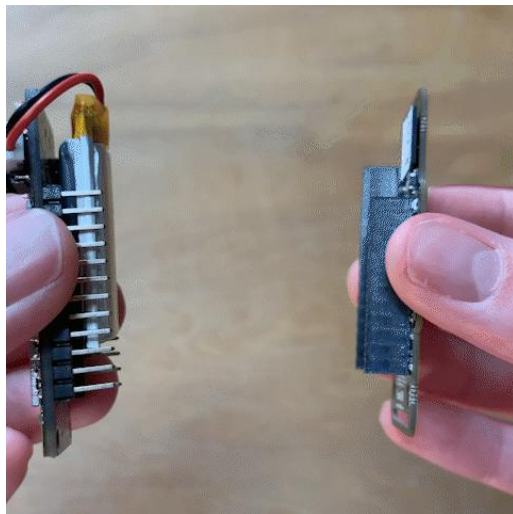


Figure 3, Depiction of the assembled boards

Software

Download

1. Install the required software based on your operating system [here](#)
2. Extract the zip file and run the .msi installer file
3. The following apps should now be available to you:

- EmotiBit Oscilloscope
- EmotiBit DataParser
- EmotiBit FirmwareInstaller

4. Make sure to enable access to public and private networks through your firewall for the Oscilloscope app

Firmware

1. Open the FirmwareInstaller and follow the steps outlined to set up the firmware in your EmotiBit
2. If you run into a problem (i.e., device is not detected), try to remove the battery and put it back
3. Finally, the EmotiBit should blink blue if it is properly connected to a network (the one specified in the config file) and the firmware is properly installed

Data Extraction

Streaming

1. Make sure that your PC is connected to the same Wi-Fi network that the EmotiBit is connected to
2. Open the Oscilloscope app, this should display the data streams for you EmotiBit
3. On 'Output List', ensure that UDP is selected
4. Now you can stream your data on any language of your choice, a simple Python listener can be seen below

```

import socket

UDP_IP = "localhost"
UDP_PORT = 12346

sock = socket.socket(socket.AF_INET,
                      socket.SOCK_DGRAM)
sock.bind((UDP_IP, UDP_PORT))

#UDP
try:
    while True:
        data, addr = sock.recvfrom(1024)
        print("received message: %s" % data)
except Exception as e:
    print("An error occurred:", e)
finally:
    sock.close()

```

Figure 4, Python listener for EmotiBit (make sure to keep Oscilloscope active)

Recording

1. To record, select record on the Oscilloscope app, and once recording is complete, unselect the button
2. Afterwards, remove the MicroSD card from the EmotiBit and place it on your PC
3. Finally, use the DataParser app to read your data

Tips

1. To conserve battery life it is important to toggle Wi-Fi off in the Oscilloscope App (check 'Power Mode' in the menu) if applicable (i.e., if not streaming in real-time)

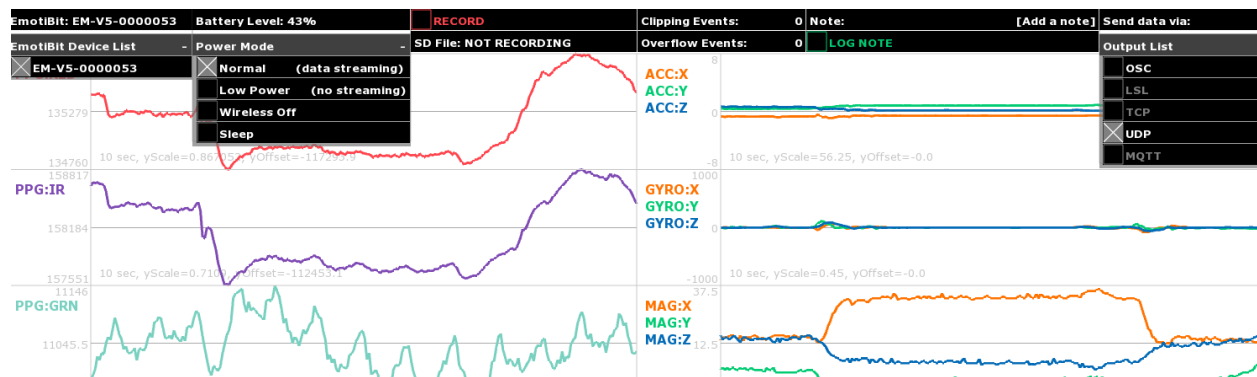


Figure 5, 'Power Mode' and 'Output List' menus in Oscilloscope

2. In between trials the device can also be put to sleep using the same function
3. To turn off the device toggle Hibernate mode on the EmotiBit board
4. To charge the device, simply connect it to your PC using the USB connector provided
5. For more detailed information visit their [GitHub repository](#)

References

- **Sensor sampling rates** (changeable)

- 25Hz - 9-axis IMU (accel, gyro, mag)
- 25Hz - PPG (3-wavelength)
- 15Hz - EDA / GSR
- 7.5Hz - Temperature

- **Size**

- 2.4in (6.1cm) length
- 1in (2.65cm) width
- 3/4in (2cm) height

- **Platforms**

- Windows
- Mac
- Linux

- **Battery life**

- 2.5 - 4 hours live-streaming
- 6 - 9 hours data-logging

- **User-owned data**

- 100% raw data + derivative metrics

- **Wireless protocols**

- WiFi
- Also possible from Adafruit: Bluetooth LoRa, 4G

- **Time-syncing with other devices**

- Typically <10msec (WiFi network dependent)

- **Open source**

- FW, SW, HW schematics
- Arduino and Adafruit Feather ecosystem compatible

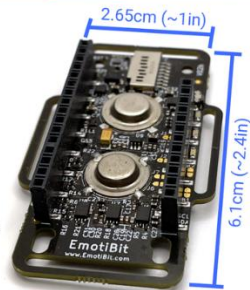


Figure 6, Specifications of the EmotiBit Device